

SEMESTER V

Course Code	Course name	L	T	P	C
CSEG3040	Cryptography and Network Security	3	0	0	3
Total Units to be Covered: 5		Total Contact Hours: 45			
Prerequisite(s):	Data Communication and Networks	Syllabus version: 1.0			

Course Objectives

1. Understand the fundamental concepts of cryptography and its applications.
2. Learn about various encryption and decryption techniques.
3. Explore network security protocols and mechanisms.
4. Develop the skills to analyze, design, and implement secure communication systems.

Course Outcomes

On completion of this course, the students will be able to

- CO1.** Relate the historical development of cryptography, its basic concepts, and terminology, and identify the different types of attacks and security services in a networked environment.
- CO2.** Illustrate the principles of symmetric-key cryptography and Public Key Cryptography, including the working of algorithms like DES, AES, and RSA and evaluate the security of block cipher modes of operation and its applications.
- CO3.** Use cryptographic hash functions, such as MD5, SHA-1, and SHA-256, for ensuring data integrity, and design applications that implement HMAC for message authentication.
- CO4.** Demonstrate an understanding of network security principles, including authentication, access control, and various network attacks, and propose countermeasures to mitigate those attacks.
- CO5.** Design and implement secure communication systems using cryptographic techniques and network security principles, and critically analyze the effectiveness of these systems in real-world scenarios.

CO-PO Mapping

Program Outcomes Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	2	2	1	-	-	-	-	-	-	-	-	-	2	-	-
CO 2	2	2	3	-	-	-	-	-	-	-	-	-	2	-	-
CO 3	2	2	3	-	-	-	-	-	-	-	-	-	2	-	-
CO 4	2	2	1	2	-	-	-	-	-	-	-	-	2	-	-
CO 5	2	2	1	2	-	-	-	-	-	-	-	-	2	-	-
Average	2	2	1.8	0.8	-	-	-	-	-	-	-	-	2	-	-

1 – Weakly Mapped (Low)

2 – Moderately Mapped (Medium)

3 – Strongly Mapped (High)

“ - ” means there is no correlation

Syllabus

Unit I: Introduction to Cryptography

5 Lecture Hours

Historical development of cryptography, Basic concepts and terminology, Types of attacks and security services, Classical cryptography and its limitations, stream and block ciphers, cryptanalysis, stereography.

Unit II: Symmetric-Key Cryptography

10 Lecture Hours

Substitution and transposition ciphers, Shannon's theory of confusion and diffusion, fiestal structure, Data Encryption Standard (DES), Triple DES, Advanced Encryption Standard (AES), Block cipher modes of operation, traffic confidentiality, key distribution, random number generation.

Unit III: Introduction to Number Theory and Public-Key Cryptography

10 Lecture Hours

Introduction to graph, ring and field, prime and relative prime numbers, modular arithmetic, Fermat's and Euler's theorem, primality testing, Euclid's Algorithm,

Introduction to public-key cryptography, RSA algorithm, Diffie-Hellman key exchange, introductory idea of Elliptic curve cryptography, Digital signatures, and certificates

Unit IV: Cryptographic Hash Functions

10 Lecture Hours

Properties and applications of hash functions, Message Digest algorithms (MD5, SHA-1, SHA-256), HMAC (Hash-based Message Authentication Code), Kerberos and X.509, directory authentication service.

Unit V: Network Security

10 Lecture Hours

Secure communication principles, Authentication and access control, Network attacks and countermeasures, Virtual Private Networks (VPNs), Transport Layer Security (TLS) and Secure Sockets Layer (SSL), Wireless Network Security, Security in Mobile and Cloud Computing

Total lecture Hours 45

Textbooks

1. William Stallings, "Cryptography and Network Security: Principles and Practice", 7th Edition, Pearson, 2017.
2. Christof Paar, and Jan Pelzl, "Understanding Cryptography: A Textbook for Students and Practitioners", Springer, 2014.

Reference Books

1. Wade Trappe, and Lawrence C. Washington, "Introduction to Cryptography with Coding Theory", 2nd Edition, Pearson, 2005.
2. William Stallings, "Network Security Essentials: Applications and Standards", 6th Edition, Pearson, 2018.

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination

Examination Scheme

Components	IA	MID SEM	End Sem	Total
Weightage (%)	50	20	30	100

Detailed breakup of Internal Assessment

Internal Assessment Component	Weightage in calculation of Internal Assessment (100 marks)
Quiz 1	15%
Quiz 2	15%
Class Test 1	15%
Class Test 2	15%
Assignment 1/Project	20%
Assignment 2/Project	20%

Course Code	Course name	L	T	P	C
CSEG3055	Formal Languages and Automata Theory	3	0	0	3
Total Units to be Covered: 4		Total Contact Hours: 45			
Prerequisite(s):	Discrete Mathematics and Design and Analysis of Algorithms	Syllabus version: 1.0			

Course Objectives

1. Understand the foundations and principles of formal languages used in computing.
2. Formulate formal and rigorous mathematical arguments.
3. Prove or disprove theorems in automata theory.
4. Develop a formal computational model pertaining to real-world problems.

Course Outcomes

On completion of this course, the students will be able to,

- CO1.** Recognize the various language classes and their relationships.
- CO2.** Comprehend regular grammar and expressions using finite automata.
- CO3.** Develop grammar and recognizer for various formal languages.
- CO4.** Analyze the decidability and intractability of different computational problems.

CO-PO Mapping

Program Outcomes Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	2	3	2	2	-	-	-	-	-	-	-	2	2	-	-
CO 2	3	3	2	2	-	-	-	-	-	-	-	2	3	-	-
CO 3	2	3	2	2	-	-	-	-	-	-	-	2	3	-	-
CO 4	2	3	2	2	-	-	-	-	-	-	-	2	2	-	-
Average	2.25	3	2	2	-	-	-	-	-	-	-	2	2.5	-	-

1 – Weakly Mapped (Low)

2 – Moderately Mapped (Medium)

3 – Strongly Mapped (High)

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Syllabus

Unit I: Finite Automata & Regular Languages

15 Lecture Hours

Computational problems vs. Formal Languages. Finite State Automata- Deterministic and Non-Deterministic, Regular Expression and Language, Closure property of Regular Language, Limitations of Regular Language, Pumping Lemma, Minimization Algorithm, Myhill-Nerode relations, Finite Automata with outputs- Mealy and Moore Machines.

Unit II: Grammar's Classification and Push Down Automata

15 Lecture Hours

Grammars and Chomsky Classification, Regular Grammar, Context Free Grammar and Languages: Ambiguity, Simplification of CFGs, Normal Forms for CFGs, Pumping Lemma for CFLs, Applications to Parsing, Pushdown Automata (PDA), PDA vs CFLs, Deterministic CFLs, Linearly Bounded Automata (LBA).

Unit III: Turing Machine

10 Lecture Hours

Introduction to Turing Machines, Configurations, Multi-tape Turing machines, Halting vs Looping, Recursive and Recursively Enumerable Languages, Decidable and Undecidable Languages.

Unit IV: Decidability & Intractability

5 Lecture Hours

Undecidability of Halting Problem, Introduction to the Theory of NP-completeness, Reductions, Rice theorem, Post Correspondence Problem, Church-Turing Thesis, Cook-Levin Theorem.

Total lecture Hours 45

Textbooks

1. John E. Hopcroft, Rajiv Motwani, and Jeffrey D. Ullman, "Automata Theory, Languages, and Computation", 3rd Edition, Pearson, 2008.
2. Dexter C. Kozen, "Automata and Computability", Springer Publishers, 2007.

Reference Books

1. Peter Linz, "An Introduction to Formal Languages and Automata", 6th edition, Jones & Bartlett India, 2016.
2. Manish Kr. Jha, "Automata Theory", 2nd Edition, S.Chand Publication, 2015.

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination

Examination Scheme

Components	IA	MID SEM	End Sem	Total
Weightage (%)	50	20	30	100

Detailed breakup of Internal Assessment

Internal Assessment Component	Weightage in calculation of Internal Assessment (100 marks)
Quiz 1	15%
Quiz 2	15%
Class Test 1	15%
Class Test 2	15%
Assignment 1/Project	20%
Assignment 2/Project	20%

Course Code	Course name	L	T	P	C
CSEG3002	Object Oriented Analysis and Design	3	0	0	3
Total Units to be Covered: 5		Total Contact Hours: 45			
Prerequisite(s):	Software Engineering	Syllabus version: 1.0			

Course Objectives

1. Learn object-oriented analysis and design and UML diagrams.
2. Apply the UML to solve several common modelling problems.
3. Learn to apply object-oriented concepts to all stages of the software development.

Course Outcomes

On completion of this course, the students will be able to

- CO1.** Describe OOAD Principles including encapsulation, inheritance, and polymorphism, and comprehend their relevance in software development.
- CO2.** Illustrate UML Modeling and the role of actors, classes, and objects in a software design
- CO3.** Illustrate activity, interaction, event, signals to design state diagram.
- CO4.** Modeling software systems using OOAD methodologies to tackle complex real-world challenges.

CO-PO Mapping

Program Outcomes Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	3	2	3	1	-	-	-	-	-	-	-	-	-	2	-
CO 2	3	2	3	1	-	-	-	-	-	-	-	-	-	2	-
CO 3	3	3	2	1	-	-	-	-	-	-	-	-	-	2	-
CO 4	3	2	2	1	-	-	-	-	-	-	-	-	-	2	-
Average	3	2.25	2.5	1	-	-	-	-	-	-	-	-	-	2	-

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Syllabus

Unit I: OOAD Basics

10 Lecture Hours

Overview of object oriented system development, Basic notion of objects, Object-oriented concepts: objects, classes, encapsulation, inheritance, polymorphism, Benefits of object-oriented analysis and design, Object basics –The unified process, Multiple Views of Objects, Modelling concepts, Modeling as a design technique, Analysis and modelling, UML and its role in software development, UML diagrams - Class diagrams, use case diagrams, Class, State, Interaction Modelling sequence diagrams, Modeling relationships between classes and objects, UML Rational Unified Process(RUP) Contrasting with Procedural Computation, Client-Server/Message Passing.

Unit II: Basic & Advanced Structural Modeling

10 Lecture Hours

Requirement Modeling: Requirement Engineering, Requirement Modeling: Use Cases Diagrams, Basic structural Modeling: Class diagrams, their components, Modeling Relationships, Common Mechanisms, Diagrams, abstract classes, Interfaces, Types and Roles, Packages. Class & object Diagrams: Terms, concepts, Modeling techniques for class & object diagrams Identifying operations and Specifying operations using CRC card.

Advanced Structural Modeling Concepts: design patterns, architectural modeling, and framework integration, Exploring advanced UML concepts like constraints, derived properties, and visibility control. Overview of C++: Procedural Extension of C, Objects, Classes and Encapsulation, Overloading, Inheritance & Polymorphism, Type Casting.

Unit III: Basic Behavioral Modeling

10 Lecture Hours

Basic Behavioral Modeling-I: Use Case Modeling, Use Case Diagrams using UML notation, use case diagram extensions and variations, Use Case Descriptions, Activity Diagrams, State Diagrams, Swim lanes Design Architecture

Basic Behavioral Modeling-II: Interaction, Interaction Diagrams, Behavioral Patterns, Scenario-Based Analysis, Verification and Validation of Behavior Models

Unit IV: Advanced Behavioral Modeling

10 Lecture Hours

Advanced Behavioral Modeling: Advanced Use Case Modeling , Activity Diagrams, State Machine Diagrams, Interaction Diagrams: Sequence and Communication Diagrams, Event-Driven Architectures, State Machines, Processes and Threads, Time and Space, State chart diagrams & Sequential Diagrams, Business Process Modeling, Modeling Reactive Systems, Advanced Behavioral Patterns, Case Studies and Practical Applications: Analyzing and modeling real-world systems using advanced behavioral modeling techniques, Case studies on event-driven systems, business processes, and reactive systems, Integration of advanced behavioral models with software development practices

Unit V: Advanced Architectural Modeling

5 Lecture Hours

Component Diagrams, Deployment Diagrams, Architectural Styles (client-server, microservices, event-driven), Architectural patterns (e.g., MVC, MVVM, layered architecture), Architectural Decision-Making: Enterprise Architecture, Distributed Systems Architecture - Case studies of distributed system architectures (e.g., cloud computing, IoT platforms), Domain-Driven Design (DDD), Service-Oriented Architecture (SOA, Event-Driven Architecture (EDA), Cloud and Serverless Architectures, Security and Privacy in Architectural Modeling, Emerging Trends and Technologies in Architecture: containerization, microservices, and serverless computing, Case studies and discussions on the practical application of new architectural approaches

Total lecture Hours 45

Textbooks

1. Michael Blaha and James Rumbaugh, "Object-oriented modeling and design with UML", 2nd Edition, Prentice-Hall of India, 2007.

2. Grady Booch, James Rumbaugh, and Ivar Jacobson, "The Unified Modeling Language User Guide", 2nd Edition, Pearson Education, 2012.
3. Russ Miles, and Kim Hamilton, "Learning UML 2.0: A Pragmatic Introduction to UML", O'Reilly, 2006.

Reference Books

1. Meilir Page-Jones, "Fundamentals of Object-Oriented Design in UML", Pearson Education, 2002.
2. Pascal Roques, "Modeling Software Systems Using UML2", WILEY- Dreamtech India Pvt. Ltd, 2004.
3. Atul Kahate, "Object Oriented Analysis & Design", The McGraw-Hill Companies, 2004.
4. John W. Satzinger, Robert B. Jackson, and Stephen D Burd, "Object-Oriented Analysis and Design with the Unified Process", Cengage Learning, 2007.
5. Hans-Erik Eriksson, Magnus Penker, Brian Lyons, David Fado, "UML 2 Toolkit", WILEY-Dreamtech India Pvt. Ltd, 2003.

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination

Examination Scheme

Components	IA	MID SEM	End Sem	Total
Weightage (%)	50	20	30	100

Detailed breakup of Internal Assessment

Internal Assessment Component	Weightage in calculation of Internal Assessment (100 marks)
Quiz 1	15%
Quiz 2	15%
Class Test 1	15%
Class Test 2	15%
Assignment 1/Project	20%
Assignment 2/Project	20%

Course Code	Course name	L	T	P	C
	Exploratory-3	3	0	0	3
Total Units to be Covered:		Total Contact Hours:			
Prerequisite(s):		Syllabus version: 1.0			

Exploratory courses offered by different schools , student has a choice to opt desired course from the available tracks .



Course Code	Course name	L	T	P	C
SLSG0205	Start your Startup	2	0	0	2
Total Units to be Covered: 10		Total Contact Hours: 30			
Prerequisite(s):		Syllabus version: 1.0			

Course Objectives

1. Foster an entrepreneurial mindset, including creativity, adaptability, and resilience, enabling students to identify and evaluate potential business opportunities!
2. Develop skills in conducting market research and competitor analysis.
3. Equip students with the tools to create comprehensive business plans and foster an understanding of the importance of effective team building and management.
4. Provide insights into funding options and strategies for securing investment.
5. Understand the challenges of scaling operations and developing sustainable growth strategies.

Course Outcomes

On completion of this course, the students will be able to:

- CO1.** Foster, identify, assess, and demonstrate an entrepreneurial mindset, including creativity, adaptability, and resilience, and enable students to identify, evaluate potential business opportunities for viability and market fit.
- CO2.** Develop skills in conducting thorough market research and competitor analysis to gather relevant data and insights.
- CO3.** Use market research findings to make informed business decisions and create a well-structured and comprehensive business plan.
- CO4.** Equip students with the tools to create comprehensive business plans and foster an understanding of the importance of effective team building and management.

CO-PO Mapping

Program Outcomes Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	3	3	2	-	-	-	-	-	-	-	3	-	-	3	3
CO 2	3	3	-	-	-	-	-	-	-	-	-	-	-	3	3
CO 3	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
CO 4	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
Average	3	3	2	1.5	-	-	-	-	-	-	.75	-	1.5	3	3

1 – Weakly Mapped (Low)

2 – Moderately Mapped (Medium)

3 – Strongly Mapped (High)

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Syllabus

Unit I:

3 Lecture Hours

Introduction to Startup world, Understanding the basics, Successful Entrepreneur's journey (can be done with alumni /seniors from Runway cohort)

Definition and characteristics of entrepreneur, motivation for starting a new startup, Exploring entrepreneurial ecosystem, entrepreneurial mindset and skills

Unit II:

3 Lecture Hours

SWOT Analysis for an Entrepreneur, challenges and overcoming challenges in Entrepreneurship.

Unit III:

3 Lecture Hours

Startup Idea Scouting Techniques and Exercises, Successful startups and validation process.

Unit IV:

3 Lecture Hours

Your target audience and validation process Part 1, Your target audience and validation process Lean Startup Methodology and MVP Development, product development and Prototyping.

Unit V:

3 Lecture Hours

Introduction to Business Model Canvas Framework, identifying key elements of startup business model Value Proposition Canvas, How to create your VPC.

Unit VI:

3 Lecture Hours

Legal considerations for startups, Intellectual Property Protection and Contracts
Sales and marketing strategies for startups Digital marketing and customer acquisition

Unit VII:

3 Lecture Hours

Managing growth and avoiding common pitfalls, scalability and Growth strategies
Importance of storytelling, Storytelling.

Unit VIII:

3 Lecture Hours

Creating a compelling pitch deck, investor's perspectives on pitching to potential investors.

Unit IX:

3 Lecture Hours

Social Impact and Sustainability in Entrepreneurship, entrepreneurship in emerging technologies and opportunities.

Unit X:

3 Lecture Hours

Review and recap, key concepts and lessons learnt, final leg Q/A, Evaluation, guidance and hand holding.

Total lecture Hours 30

Textbooks

1. Entrepreneurship: The Practice and Mindset Start Your Own Business.

Reference Books

1. Simon Sinek - The Infinite Game

2. Eric Ries - The Lean Startup
3. Chris Gillebeau - The \$100 Startup
4. Simon Sinek - Start With Why
5. Peter Thiel - Zero to One

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination

Examination Scheme

Component	QUIZ	Pitch Weightage	Total
Weightage (%)	60	40	100

Course Code	Course name	L	T	P	C
CSEG3060	Research Methodology in CS	3	0	0	3
Total Units to be Covered: 06		Total Contact Hours: 45			
Prerequisite(s):		Syllabus version: 2.0			

Course Objectives

1. To Familiarize Students with Research Problem Identification
2. To Introduce Various Research Methodologies in Computing Science
3. To Promote Ethical Conduct in Research
4. To Enhance Technical Writing and Proposal Development Skills
5. To Provide Insights into Intellectual Property and Patents

Course Outcomes

On completion of this course, the students will be able to

- CO1.** Identify and formulate research problems using defined criteria and characteristics.
- CO2.** Utilize various investigative approaches and methodologies for problem-solving in computing science research.
- CO3.** Demonstrate ethical conduct and awareness of research ethics and plagiarism in their work.
- CO4.** Produce technically sound and well-structured research reports and proposals.
- CO5.** Analyze and apply knowledge of intellectual property rights, particularly patents, in the context of computing science.

CO-PO Mapping

Program Outcomes Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	-	2	2	3	-	-	-	-	-	-	-	2	-	-	-
CO 2	-	2	2	3	-	-	-	-	-	-	-	2	-	-	-
CO 3	-	2	2		-	-	-	-	-	-	-	2	-	-	-
CO 4	-	-	2	-	-	-	-	3	-	2	2	2	-	-	-
CO 5	-	-	2	-	-	-	-	2	-	2	2	2	-	-	-

Average	-	1.2	2	1.2	-	-	-	1	-	.8	.8	2	-	-	-
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1 – Weakly Mapped (Low)

2 – Moderately Mapped (Medium)

3 – Strongly Mapped (High)

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Syllabus

Unit I: Introduction to Research Problem

9 Lecture Hours

Understanding Research Problem: Meaning and Significance, Sources and Identification

Criteria and Characteristics of a Good Research Problem: Essential Attributes, Errors in Selection ***Scope, Objectives, and Approaches:*** Defining Scope and Objectives, Investigative Approaches for Solutions, Data Collection, Analysis, Interpretation, Necessary Instrumentation.

Unit II: Research Methods in Computing Science

13 Lecture Hours

Dialectic of Research in Computing Science: Models of Argument, Proof Methods (Demonstration, Empirical, Mathematical)

Deduction and Induction for Computer Science: Theoretical Models and Approaches, Algorithmic and Software Engineering Approaches, Mathematical Modelling, Performance Estimation and Evaluation

Unit III: Ethical Conduct and Literature Studies

8 Lecture Hours

Effective Literature Studies: Approaches and Analysis, Addressing Plagiarism,

Research Ethics: Ethical Considerations in Research.

Unit IV: Technical Writing and Research Proposal Development 5Lecture Hours

Effective Technical Writing, Report Writing Techniques, Developing a Research Proposal, Proposal Format and Components, Presentation and Review Committee Assessment.

Unit V: Intellectual Property and Patents

5 Lecture Hours

Nature and Process of Intellectual Property: Overview of Patents, Designs, Copyrights, Patenting Process: Research, Innovation, Development

International Scenario and Patenting: International Cooperation on Intellectual Property

Procedures for Patent Grants and Patenting under PCT

Unit VI: Patent Rights and New Developments in IPR

5 Lecture Hours

Scope and Transfer of Patent Rights: Licensing and Technology Transfer, Patent Information and Databases

New Developments in Intellectual Property Rights: Administration of Patent System, IPR in Computer Software, etc., Case Studies and IPR in Educational Institutes.

Total lecture Hours 45

Textbooks

1. "Research methodology: an introduction for science & engineering students", Stuart Melville and Wayne Goddard
2. Research methodology for engineers by R Ganesan, MJP Publishers

Reference Books

1. Algorithm Design Manual by Steven Skiena, Springer publication.

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination

Examination Scheme

Components	IA	MID SEM	End Sem	Total
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Weightage (%)	50	20	30	100
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Detailed breakup of Internal Assessment

Internal Assessment Component	Weightage in calculation of Internal Assessment (100 marks)
Quiz 1	15%
Quiz 2	15%
Class Test 1	15%
Class Test 2	15%
Assignment 1/Project	20%
Assignment 2/Project	20%



Course Code	Course name	L	T	P	C
CSEG3056	Probability, Entropy, and MC Simulation	3	0	0	3
Total Units to be Covered: 6		Total Contact Hours: 45			
Prerequisite(s):	Advanced Engineering Mathematics – 1 & 2, Discrete Mathematics, Linear Algebra	Syllabus version: 1.0			

Course Objectives

This course envisions to impart into the students to:

1. Develop a solid foundation in probability, enabling them to recognize probability distributions in real-world scenarios and effectively address probabilistic problems.
2. Equipping with the necessary skills to proficiently utilize Monte Carlo algorithms in tackling intricate problems that involve uncertainty and randomness.
3. Enhances students' analytical and computational abilities to tackle practical applications within the realm of computer science.
4. Possess the requisite knowledge and tools to confidently approach real-world problems that demand expertise in probabilistic analysis, information processing, and simulation techniques.

Course Outcomes

On completion of this course, the students will be able to

- CO1.** Identify appropriate probability distributions, discrete / continuous, in problems with uncertainty and compute quantities of interest.
- CO2.** Analyze multivariate data and draw statistical information by identifying the appropriate random variate.
- CO3.** Gain understanding of probabilistic behavior of complex uncertain systems through Monte-Carlo simulation experiments.
- CO4.** Calculate uncertainty in a random experiment using entropy function and learn to compute the distance between two probability distributions using KL divergence.

CO-PO Mapping

Program Outcomes Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	2	2	1	-	-	-	-	-	-	-	-	1	2	1	-
CO 2	2	2	1	-	-	-	-	-	-	-	-	1	2	1	-
CO 3	2	2	1	2	1	1	-	-	-	-	-	1	2	1	-
CO 4	2	2	1	2	1	1	-	-	-	-	-	1	2	1	-
Average	2	2	1	1	.5	.5	-	-	-	-	-	1	2	1	-

1 – Weakly Mapped (Low)

2 – Moderately Mapped (Medium)

3 – Strongly Mapped (High)

“ - ” means there is no correlation

Syllabus

Unit 0: Motivation

1 Lecture Hours

Randomness/Uncertainty all pervasive, Probability theory gives rational description while entropy provides quantitative measure of uncertainty, Generation of random numbers for conducting numerical experiments for simulations.

Unit I: Probability

8 Lecture Hours

Motivation: Frequency based probability, Probability models, Sample space, Algebra of events, Notion of statistical regularity, Simpson's Paradox, Axioms of probability, Some probability propositions, Conditional probability, Independence of events, Birthday problem, How to make a fair coin from a biased coin?, Reliability: Series and parallel systems, Theorem of total probability, Bayes' theorem, Debate: Frequentist vs. Bayesian, Bernoulli trials, Geometric probability – continuous sample space. Recap of Unit I.

Unit II: Discrete Random Variables

8 Lecture Hours

Definition of random variables, Distribution function, Discrete RV, Probability mass function, Expectation, Mean and variance, Moment generating function. Special discrete distributions: Bernoulli and Binomial, Poisson, Geometric and memoryless property of Geometric pmf, Negative binomial, Hypergeometric, Uniform. Indicator random variable. Recursive formulation for binomial probabilities. Recap of Unit II.

Unit III: Continuous Random Variables

9 Lecture Hours

Definition, Distribution function, Probability density function, Expectation, Mean and variance, Moment generating function. Special continuous distributions: Uniform, Exponential, Gamma, Normal, Pareto. Weibull distribution, Reliability, and failure rate. Functions of random variable. Markov inequality and Chebyshev's inequality. Limiting distributions and Stirling's approximation. Recap of Unit III.

Unit IV: Jointly Distributed Random Variables

7 Lecture Hours

Random vectors, Joint probability distribution, Independent random variables, Sum of independent random variable, Conditional distribution, Conditional expectation, Expected number of comparisons in Quick sort, Covariance, Correlation coefficient, Multivariate normal distribution. Limit theorems: Law of large numbers and Central limit theorem, Breakdown of CLT. Recap of Unit IV.

Unit V: Monte-Carlo (MC) Simulation

7 Lecture Hours

MC methods – sampling and simulation, Analogy between probability and volume, Estimation of π , Buffon's needle problem, Estimation of integrals, Linear congruential methods, Pseudo random numbers, Random variable generation - Inverse transform method, Acceptance rejection method. Generating continuous random variables: Exponential, Normal - Box-Muller approach, Gamma distribution. Generating discrete random variables: Binomial and Poisson variables. Illustration of CLT using random numbers. MC simulation of some probability models. Recap of Unit V.

Unit VI: Information Theory and its Applications

5 Lecture Hours

Overview of information theory, Information / Surprise and Entropy, Entropy as a measure of randomness, Properties of Entropy function, Applications of Entropy: English language, music, logical problems, lossless compression. Kullback-Leibler measure of divergence. Recap of Unit VI.

Total lecture Hours 45

Textbooks

1. S. M. Ross, "A first course in probability", Pearson, 2020.
2. K. S. Trivedi, "Probability & statistics with reliability, queuing and computer science applications", John Wiley & Sons, 2008. Chapters 1 to 5.
3. R. Y. Rubinstein, and D. P. Kroese, "Simulation and the Monte Carlo method", John Wiley & Sons, 2016.

Reference Books

1. M. Mitzenmacher, and E. Upfal, "Probability and computing: Randomization and probabilistic techniques in algorithms and data analysis", Cambridge University Press, 2017.
2. R. Nelson, "Probability, stochastic processes, and queueing theory: the mathematics of computer performance modeling", Springer Science & Business Media, 2013.
3. I. Goodfellow, Y. Bengio, and A. Courville, "Deep learning", MIT press, 2016. Chapter 3.

Modes of Evaluation: Quiz/Assignment/ presentation/ extempore/ Written Examination

Examination Scheme

Components	IA	MID SEM	End Sem	Total
Weightage (%)	50	20	30	100

Detailed breakup of Internal Assessment

Internal Assessment Component	Weightage in calculation of Internal Assessment (100 marks)
Quiz 1	15%
Quiz 2	15%
Class Test 1	15%
Class Test 2	15%
Assignment 1/Project	20%
Assignment 2/Project	20%



Course Code	Course name	L	T	P	C
EMPL003	EDGE – Advance Communication	1	0	0	0
Total Units to be Covered:		Total Contact Hours:			
Prerequisite(s):		Syllabus version: 1.0			

Employment training related course

